A One-Loop BV Obstruction in Pure Weyl Gravity and Its Formal Tau-Adic Wess–Zumino Resolution

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Abstract

We compute the coefficient-bearing one-loop BV anomaly of strict four- dimensional pure Weyl gravity in a complete local gauge-fixed quotient. The repository Euclidean elliptic complex, measure, zero-mode, contour, parity, and local heat-kernel ledgers give

$$\mathcal{A}^{(1)} = \frac{1}{(4\pi)^2}\left(\frac{199}{30}[\omega C^2] - \frac{87}{20}[\omega E_4]\right).$$

Both coordinates are nonzero in the strict quotient. Hence the fixed-field- content local Euclidean quantum master equation is obstructed at one loop. An exact dual-cone witness further excludes cancellation by any nonnegative combination of standard-sign free conformal scalars, Weyl or Dirac fermions, and gauge vectors.

We then adjoin a shifting scalar τ and derive its complete minimal-BV cotangent lift from $S_{\tau,\min} = \int \tau^*(\mathcal{L}_\xi \tau + \omega)$. In dressed canonical variables

$$\widehat{g} = e^{-2\tau} g, \qquad \widehat{g}^* = e^{2\tau} g^*, \qquad \widehat{\tau}^* = \tau^* + 2 g_{\mu\nu} g^{*\mu\nu},$$

the Weyl sector is an exact contractible quartet. In the formal tau-adic local analytic jet algebra the complete dimension-four extended quotient is

$$\dim H^{0,4}_{\mathrm{ext}} = (3\ \mathrm{even}, 1\ \mathrm{odd}), \qquad H^{1,4}_{\mathrm{ext}} = 0.$$

The coefficient-bearing Wess–Zumino counterterm therefore restores the extended local Euclidean QME at one loop.

The two dispositions refer to different theories and are not contradictory: strict field content is obstructed, while the formal tau-adic compensator extension is locally restored at one loop. Neither statement is an all-loop or Lorentzian QME theorem. No Hadamard state, particle Hilbert space, unitarity theorem, or residual quantum transfer is claimed.

AI-use disclosure and computational accountability

OpenAI GPT-5 was used substantively for mathematical synthesis, verification code, claim auditing, and manuscript drafting. The human author directed the programme and remains responsible for the claims, citations, and final text. All cohomological ranks and cancellations reported here use exact rational arithmetic. Every headline statement is bound to a deterministic certificate and an independent verifier. Submission remains conditional on final human and literature review.

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Contents

1 Question, result, and claim boundary

Pure Weyl gravity has classical Diff×Weyl gauge symmetry. The quantum question is not whether a determinant has a nonzero logarithm, but whether the regularized Slavnov breaking is trivial in the relative BV cohomology $H^{1,4}(s \mid \mathrm{d})$. We therefore separate three calculations:

1. classify the complete local obstruction space;
2. compute the coefficient vector in one specified analytic route;
3. decide whether that vector is a boundary in the chosen field theory.

The result has two theory labels which must not be suppressed:

Theory	One-loop local Euclidean disposition
strict pure Weyl gravity, fixed field content	obstructed
formal tau-adic Wess–Zumino compensator extension	QME restored

The second row does not reverse the first. It changes the BV complex.

All local results carry LOCAL-ALGEBRAIC; the coefficient computation also carries EUCLIDEAN-SPECTRAL. No result in this paper carries LORENTZIAN-CAUSAL.

2 Strict local BV obstruction space

The local jet algebra tracks ghost number, antifield number, form degree, engineering dimension, Grassmann parity, spacetime parity, and Weyl weight. Tensor symmetries, algebraic and differential Bianchi identities, graded commutativity, dummy-index relabelling, integration by parts, and total derivatives are reduced before exact sparse linear algebra is performed.

On the regular Bach-locus chart, the minimal Koszul–Tate sector contracts at positive antifield number. Ten nonminimal pairs contract explicitly, and an arbitrary invertible local BV-canonical gauge-fixing transformation preserves the quotient. The pure-diffeomorphism and mixed Diff–Weyl total complexes add no independent four-dimensional anomaly class.

Theorem 2.1 (Strict dimension-four anomaly quotient)**.** *In the declared complete gauge-fixed local BV complex,*

$$\dim H^{1,4}_{\text{even}}(s \mid \mathrm{d}) = 2, \qquad \dim H^{1,4}_{\text{odd}}(s \mid \mathrm{d}) = 1.$$

Representatives may be chosen as

$$[\omega C^2], \qquad [\omega E_4], \qquad [\omega C\widetilde{C}].$$

The type-D carrier $\omega\Box R$ is exact with its stored R^2 counterterm and horizontal current.

Closure, exact primitives, complete basis manifests, and normalized dual nonmembership witnesses are stored independently. The universal diffeomorphism descent and intrinsic Weyl descent are recorded separately; in particular, a nonzero Diff-horizontal tower is not called a nontrivial intrinsic Weyl descent.

This separation matches the algebraic Weyl-anomaly classification of Boulanger [3]: the Euler/type-A class is distinguished by a nontrivial intrinsic Weyl descent, whereas strictly Weyl-invariant type-B densities have trivial intrinsic Weyl descent. The present calculation has a different target and scope. It retains the universal Diff completion, the dynamical metric antifields, the Koszul–Tate filtration, and the explicit minimal-to-gauge-fixed BV comparison needed for the pure-gravity master equation.

3 Coefficient-bearing one-loop breaking

The coefficient route combines two independent curvature carriers. A local Ricci-flat carrier exposes C^2, while round S^4 fixes the Euler coordinate. The complete Euclidean symbol sequence has bundle dimensions $5 \to 10 \to 5$, exact image–kernel equality, a formal adjoint, and the required Diff/Weyl nonminimal doublets. Determinant multiplicities, the functional measure, conformal-Killing zero modes, the indefinite contour, and the parity Ward identity are included in the accepted integration slice.

Theorem 3.1 (Repository one-loop Slavnov breaking)**.** *In the normalization*

$$S_W = \alpha_C \int \sqrt{g}\, C^2,$$

the repository-regulated breaking has coordinates

$$(C^2, E_4, C\widetilde{C}, \Box R) = \left(\frac{199}{30}, -\frac{87}{20}, 0, 0\right).$$

After reduction against the complete strict quotient,

$$\mathcal{A}^{(1)} = \frac{1}{(4\pi)^2}\left(\frac{199}{30}[\omega C^2] - \frac{87}{20}[\omega E_4]\right).$$

Consequently the strict fixed-field-content local Euclidean QME is obstructed at one loop.

The odd zero is derived from the parity Ward identity for the declared real tensor-Laplacian regulator. The type-D zero is scheme-dependent but exact; it cannot cancel either nontrivial coordinate. A beta function alone would not prove this theorem: the decisive object is the regulated Slavnov breaking reduced in the complete BV quotient.

The even numbers agree, after translating the Euler-sign convention, with the classic one-loop Weyl-gravity anomaly of Fradkin and Tseytlin [4]. Our claim is not that these constants were previously unknown. It is the content-addressed identification of the repository elliptic complex, multiplicities, measure, zero modes, contour, and regulator with a complete BV obstruction basis, followed by the explicit Slavnov-class disposition. The local $\Box R$ term has a separate scheme-sensitive history; the recent heat-kernel analysis of Barvinsky et al. [6] makes that distinction explicit. In the scheme frozen here its coefficient is zero and the class is in any case removed by the stored R^2 primitive.

Proposition 3.2 (Standard unitary free matter cannot cancel the vector)**.** *Let the added matter content be any nonnegative combination of real conformal scalars, Weyl or Dirac fermions, and standard gauge vectors. No such combination cancels both even coordinates of the strict breaking.*

The proof is an exact separating dual-cone witness. Interacting fixed points, wrong-sign or nonunitary matter, conformal higher spins, and compensator extensions lie outside this no-go.

4 The compensator cotangent lift

Adjoin a scalar with BRST transformation

$$Q\tau = \mathcal{L}_\xi \tau + \omega.$$

Its cotangent rows are not guessed. They are the Euler derivatives of the single minimal master term

$$S_{\tau,\min} = \int \tau^*(\mathcal{L}_\xi \tau + \omega).$$

In addition to the field row this forces

$$\delta\omega^* = 2g_{\mu\nu}g^{*\mu\nu} + \tau^*, \qquad \delta\xi^*_\mu = N_{\xi,\mu} + \tau^*\nabla_\mu\tau, \qquad \gamma\tau^* = \mathcal{L}_\xi\tau^*,$$

together with the opposite-adjoint Lie-prolonged rows.

Theorem 4.1 (Exact minimal-BV cotangent lift)**.** *On every strict-plus-compensator atom,*

$$\delta^2 = 0, \qquad \delta\gamma + \gamma\delta = 0, \qquad Q^2 = 0.$$

The canonical change

$$\widehat{g} = e^{-2\tau}g, \quad \widehat{g}^* = e^{2\tau}g^*, \quad \widehat{\tau}^* = \tau^* + 2g_{\mu\nu}g^{*\mu\nu}$$

preserves the canonical one-form and sends the Weyl Koszul–Tate row to $\delta\omega^* = \widehat{\tau}^*$. Thus $(\tau, \omega, \omega^*, \widehat{\tau}^*)$ is an exact contractible quartet.

The exponential change is an automorphism only after declaring the formal tau-adic local analytic completion. Its two inverse series are replayed with exact rational coefficients and obey at every order

$$\sum_{k=0}^{n} \frac{(-2)^k}{k!}\frac{2^{n-k}}{(n-k)!} = \frac{2^n}{n!}(1-1)^n = \delta_{n0}.$$

No finite-polynomial-in-τ theorem is inferred.

5 Extended cohomology and QME restoration

Quartet contraction reduces the extended complex to pure Diff BV cohomology in the dressed metric $\widehat{g}$. The dimension-four counterterm inventory is therefore larger than in strict Weyl gravity: $R(\widehat{g})^2$ is now closed.

Theorem 5.1 (Formal tau-adic extended quotient)**.** *In the declared formal tau-adic local analytic jet algebra,*

$$H^{0,4}_{\mathrm{ext,even}} = \mathrm{span}\{C(\widehat{g})^2, E_4(\widehat{g}), R(\widehat{g})^2\},$$
$$H^{0,4}_{\mathrm{ext,odd}} = \mathrm{span}\{C(\widehat{g})\widetilde{C}(\widehat{g})\}, \qquad H^{1,4}_{\mathrm{ext}} = 0.$$

$\Box R(\widehat{g})$ remains horizontally exact. There is no incoming dimension-four positive-antifield boundary: the only dimension-zero natural symmetric tensor is $\widehat{g}$, whose image is the vanishing trace of the Bach row.

Explicit primitives for the strict four-candidate anomaly carrier are

$$B_C = \int \sqrt{g}\,\tau C^2, \qquad B_P = \int \sqrt{g}\,\tau C\widetilde{C},$$

together with the Euler Wess–Zumino functional B_E and the stored R^2 primitive $B_\Box$. Their boundary matrix has exact rank four. The curved-background Euler primitive agrees with the standard dilaton Wess–Zumino construction obtained from finite Weyl transformation of dimensionally regulated counterterms [5]. The new certificate needed here is not the displayed AFN0 formula alone: it is the cotangent lift, quartet contraction, exhaustive extended $H^{0,4}/H^{1,4}$ comparison, and coefficient-bearing BV boundary equation.

Corollary 5.2 (One-loop local Euclidean QME restoration)**.** *The counterterm*

$$-\frac{\hbar}{(4\pi)^2}\left(\frac{199}{30}B_C - \frac{87}{20}B_E\right)$$

satisfies

$$\mathcal{A}^{(1)} + s_{\mathrm{ext}}S^{(1)}_{\mathrm{ct}} = 0 \pmod{\mathrm{d}}.$$

Hence the compensator-extended local Euclidean QME is restored at one loop in the declared formal tau-adic algebra.

Proposition 5.3 (The restored QME does not determine a unique Q_1)**.** *The local Wess–Zumino Hamiltonian contribution*

$$Q_1^{\mathrm{WZ}} = (C_1^{\mathrm{WZ}}, \cdot)$$

is coefficient-bearing and fixed by the counterterm above. The complete renormalized first Slavnov operator is nevertheless underdetermined. On the flat Euclidean fixtures $p = (1,0,0,0)$ with one transverse-traceless and one conformal polarization, the bulk quadratic response of the allowed basis $(C(\widehat{g})^2, E_4(\widehat{g}), R(\widehat{g})^2, C(\widehat{g})\widetilde{C}(\widehat{g}))$ is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 \end{pmatrix}.$$

Thus the C^2 and R^2 finite-counterterm directions change the bulk Hamiltonian vector field independently. This response calculation alone does not supply a nonlocal representative, the renormalized BV Laplacian or time-ordered product, or finite normalization conditions. The next proposition supplies the anomaly-induced nonlocal component, but not the Weyl-invariant finite remainder.

Proposition 5.4 (Anomaly-induced Euclidean representative)**.** *Set*

$$\mathcal{E}_4 = E_4 - \frac{2}{3}\Box R, \qquad \Delta_4 = \Box^2 + 2R^{\mu\nu}\nabla_\mu\nabla_\nu - \frac{2}{3}R\Box + \frac{1}{3}(\nabla^\mu R)\nabla_\mu.$$

On an invertible Euclidean boundary problem, or on the compatible source sector of a self-adjoint generalized inverse G_4 of Δ_4, one anomaly-induced representative is

$$\Gamma_{1,\mathrm{anom}} = \frac{1}{(4\pi)^2}\left\{\frac{1}{8}\left\langle \mathcal{E}_4, G_4\left(2cC^2 - a\mathcal{E}_4\right)\right\rangle + \frac{a}{18}\int \sqrt{g}\, R^2\right\}, \qquad (c,a) = \left(\frac{199}{30}, \frac{87}{20}\right).$$

Its Weyl variation is the certified anomaly $(4\pi)^{-2}(cC^2 - aE_4)$ in that sector. The three functional coefficients are

$$\left(\frac{199}{120}, -\frac{87}{160}, \frac{29}{120}\right).$$

The last coefficient follows from $\delta_\sigma \int \sqrt{g}R^2 = -12\int\sqrt{g}\,\sigma\Box R$ and is required to return from the canonical $\mathcal{E}_4$ representative to the repository $\Box R = 0$ convention. This is the Riegert construction [7] with the scheme conversion made explicit; it is not the complete finite effective action.

6 What does not follow

The strict obstruction and extended restoration are local one-loop results. They do not establish:

- an all-loop extended QME;
- a Lorentzian renormalized time-ordered product or Slavnov identity;
- a BRST-compatible global Hadamard state;
- positivity, a particle Hilbert space, scattering, or unitarity;
- survival or removal of either residual deformation class;
- a quantum Cartan identity or residual moment-map algebra.

Residual transfer is specifically forbidden at present. The frozen classical contraction belongs to the strict theory and has no compensator rows. The local Wess–Zumino contribution to Q_1 is fixed, but the proposition proves that it is not a unique complete renormalized operator. The anomaly-induced Paneitz/Riegert representative fixes one nonlocal solution, but its Weyl-invariant finite remainder, normalization, global Green/kernel data, and renormalized-product contribution remain open. The next gate is a compensator-inclusive classical contraction together with those missing data. Bridge 4 also remains inactive until a normalized same-background classical mode, causal Green carrier, and pairing coexist. Bridge 5 has a QME disposition but lacks the independent interaction-to-physical-branch map. No classical tangent obstruction is therefore called ghost removal or a quantum constraint.

7 Machine certification and reproduction

The two theorem receipts are

- `quantum-weyl/anomalies/certificates/REGULATED_REPOSITORY_BV_SLAVNOV_BREAKING.json`;

- quantum-weyl/anomalies/certificates/WESS_ZUMINO_EXTENDED_LOCAL_BV_COHOMOLOGY.json.
- quantum-weyl/transfer/certificates/ONE_LOOP_SLAVNOV_Q1_DISPOSITION.json.
- quantum-weyl/transfer/certificates/ANOMALY_INDUCED_NONLOCAL_GAMMA1.json.

The cotangent-lift and matter-cone receipts are separate dependencies. The companion computational supplement prints the exact quotient ranks, orbit counts, coefficient vector, quartet matrices, extended boundary matrix, and input-hash ledger from generated certificate tables. The claim map binds the manuscript, both PDFs, the generated tables, their producer and verifier, and all explicit nonclaims.

The scoped replay is:

```
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.verify_wess_zumino_minimal_bv_cotangent_lift
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.verify_wess_zumino_extended_local_bv_cohomology
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_one_loop_slavnov_q1_disposition
PYTHONPATH=quantum-weyl python3 -m verify_active_frontier
PYTHONPATH=quantum-weyl python3 -m atlas.verify_quantum_atlas_fragment
python3 paper/verify_12_pure_weyl_one_loop_bv_anomaly_claim_map.py
```

8 Outlook

The immediate local-to-residual question is now sharp. Supply finite normalization conditions, the finite nonlocal effective action and renormalized BV operator data to fix a complete Q_1; then, given an extended contraction $(\iota_{\mathrm{ext}}, \pi_{\mathrm{ext}}, S_{\mathrm{ext}})$, compute

$$q_1 = \pi_{\mathrm{ext}} Q_1 \iota_{\mathrm{ext}}$$

and test first-order nilpotency, the quantum Cartan defect, incoming $H^3 \to H^4$, outgoing $H^4 \to H^5$, operator mixing, and the corrected pairing. This is a new theorem, not a corollary of local QME restoration.

The independent Lorentzian rail must construct renormalized causal products and a compatible state before any unitary or particle conclusion is possible. The present result nevertheless fixes the finite local obstruction vector that every such construction must reproduce or trivialize.

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